1.Missing Values

- Are there any missing or null values in the dataset?
- How would you handle missing data — delete rows, fill with mean/median, or use other imputation methods?

Ans: I checked for missing values using the COUNTBLANK function in Excel, and I found that there were no missing or null values in the dataset. In general, if a column has too many missing values and does not add much value to the analysis, I would prefer to drop that column. If only a few values are missing and the column is important for analysis, I would impute the missing values using the mean or median.

2.Data Formatting

- Are all column names clean and properly formatted (no spaces, consistent casing)? - Yes
- Are the data types (numeric, categorical, date) correctly assigned? -No. I Assigned the appropriate data type

3.Duplicate Handling

- Are there any duplicate records in the dataset?
- If yes, how would you identify and remove them?

Using Conditional Formatting, checked for duplicate values.

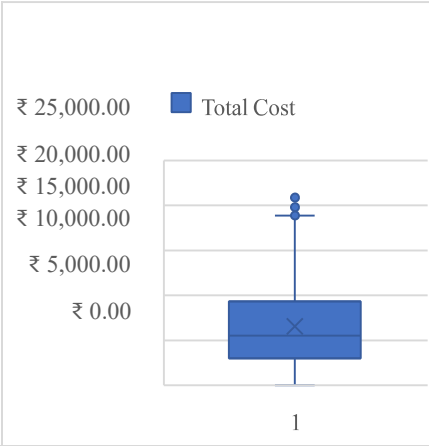
If there I any duplicate row, to remove that I used Remove Duplicates in Data field.

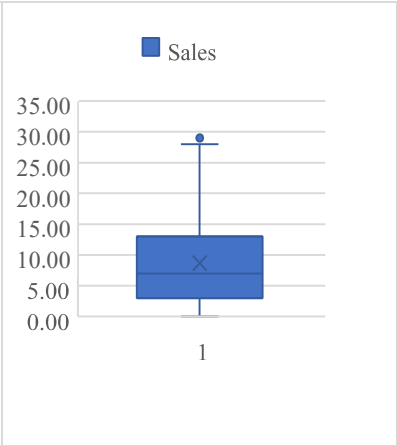
4.Outlier Detection

- How can you identify outliers using Excel formulas or conditional formatting?
- Should outliers be removed or treated?

Ans: I identified outliers using a box plot in Excel, where values outside the whiskers were marked as outliers. I observed that the Sales and Total Cost columns contain outliers. Since this is sales data, the outliers likely represent genuine business scenarios such as high-value sales rather than data errors. The Total Cost column shows outliers because it is directly dependent on the Sales column. Therefore, these outliers do not need to be removed.

Handling Outliers: Using IQR method





5.Feature Understanding

- What does each column represent?

Ans: Column are Campaign, Total Leads, Sales, Conversion Rate (%), Cost per Lead and Total Cost.

Campaign: Name of Marketing Campaign

Total Leads: Number of Potential Customers who showed interest

Sales: Number of Customers converted from leads

Conversion Rate (%): Percentage of Leads converted into Customer s

Cost per Head: Cost to get each lead

Total Cost: Total cost of the campaign

- Are there categorical columns that need to be encoded (e.g., "Yes/No", "Male/Female")?

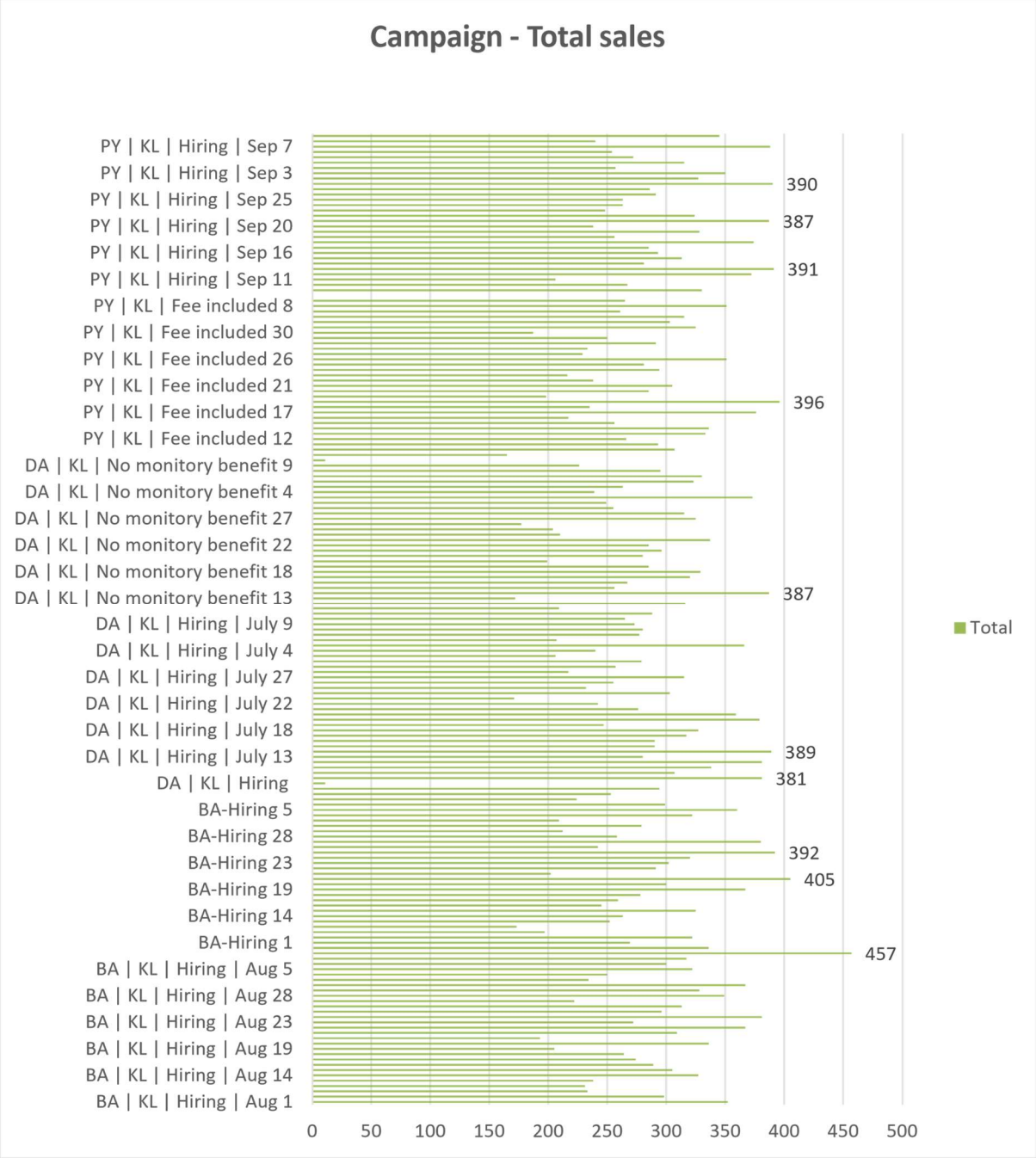
Ans: No, the Campaign column is the only categorical variable in the dataset, and it is not used for prediction.

6.Descriptive Statistics

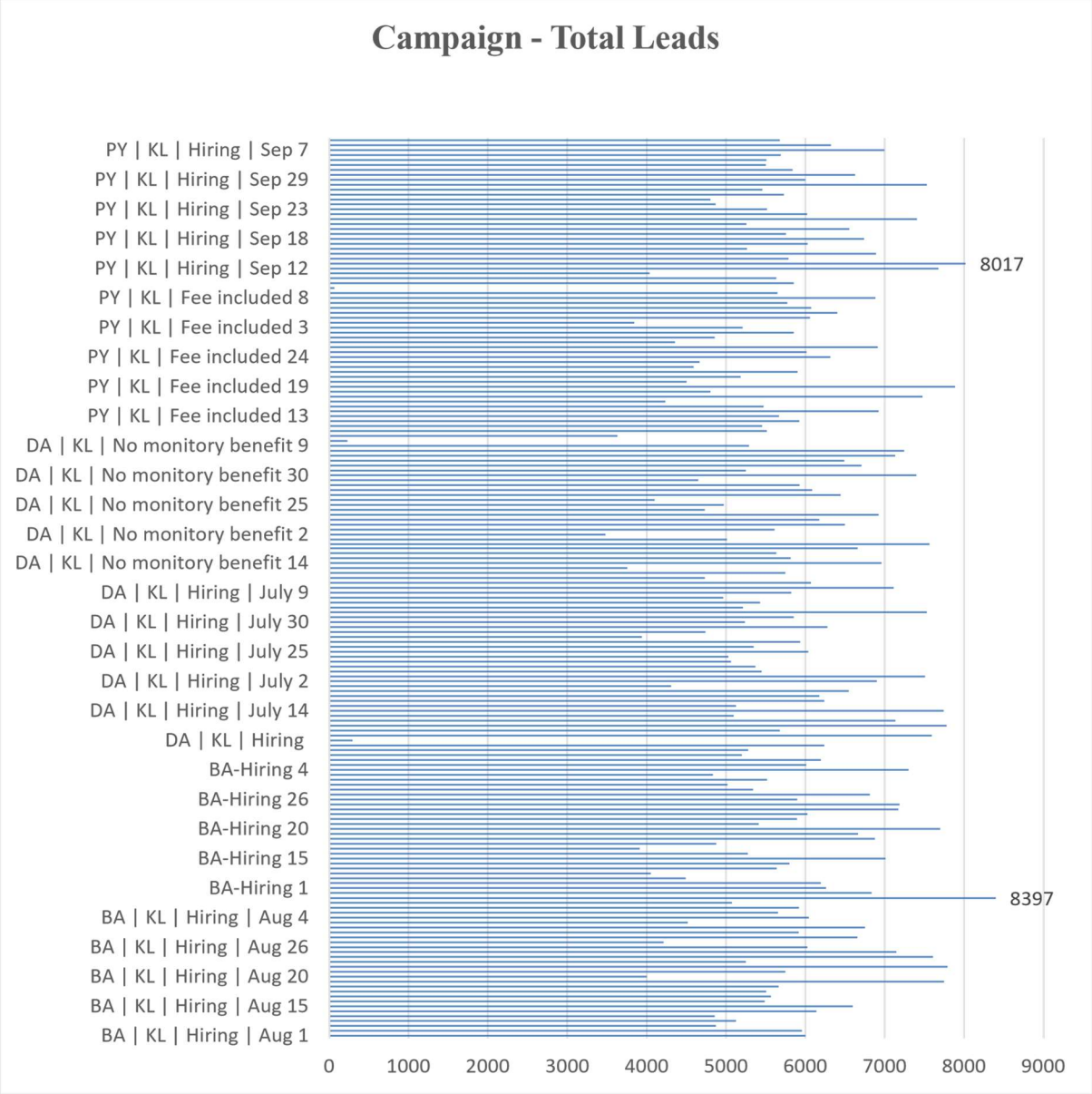
- Find average, median, and standard deviation for key numeric features.

Average	176.3511667	8.654833333	4.876548333	37.22025333	6558.923513
Median	178	7	4.91	37.35	5533.065
Std	72.03660686	6.679081398	300.27%	18.71297781	4476.376769

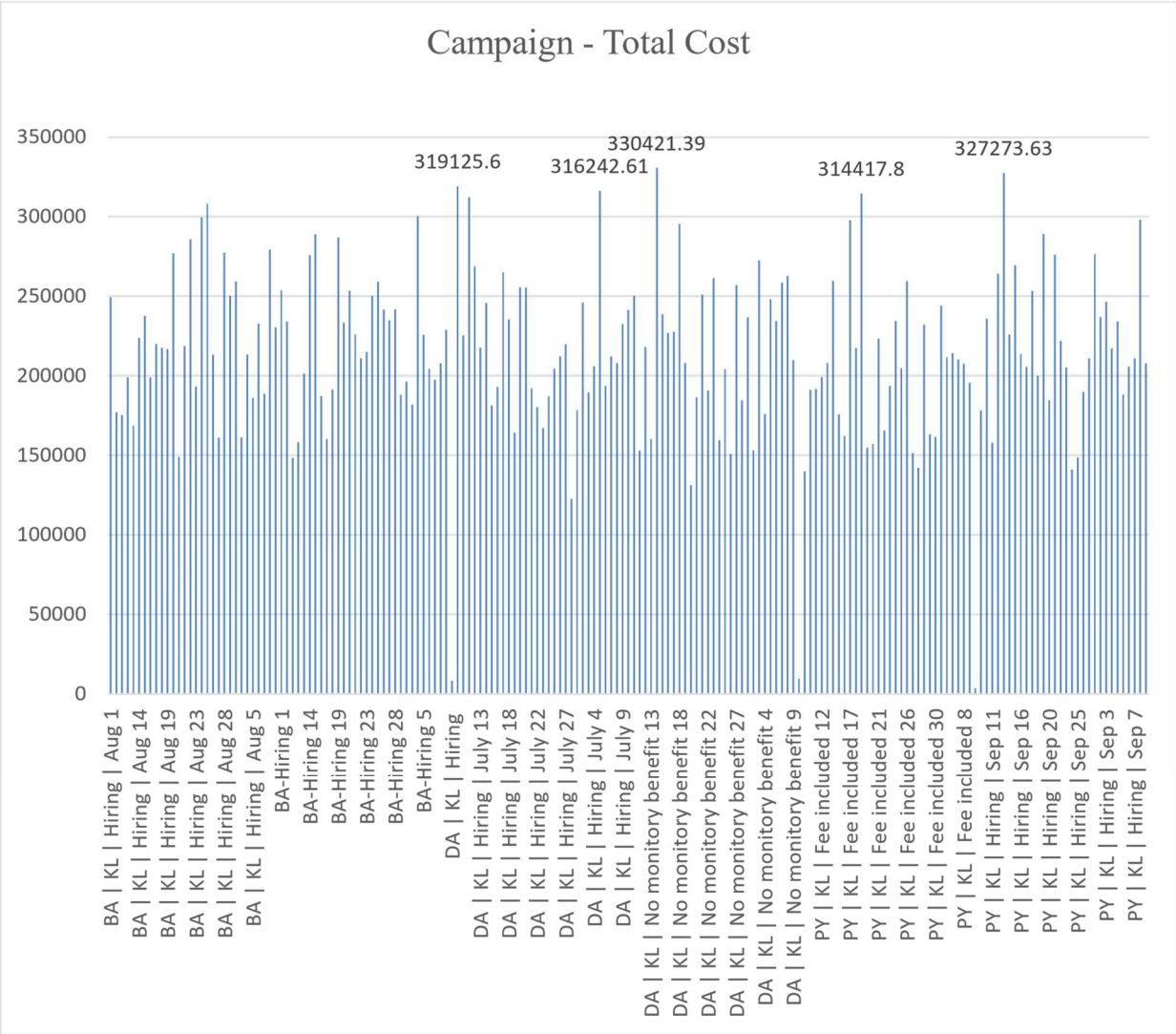
- What insights can you draw from pivot tables or charts (e.g., campaign success by age group, or by region)?



- Highest Sales - BA | KL | Hiring | Aug 8
- Other Strong Campaigns: BA-Hiring 20 - 405
- Most of the Sales is from Hiring Campaign



- Highest Leads - BA | KL | Hiring | Aug 8
- Other Highest Leads: BA-Hiring 4 with 8017



Highest Cost-DA | KL | No monitory benefit 14
Other Campaigns with Highest Cost: DA | KL | Hiring | July 1, DA | KL | Hiring | July 5, and PY | KL | Fee included 19

<u>Conclusion</u>

More leads usually increase the chance of sales

Cost per Lead affects number of leads and campaign efficiency

ML

1.Target Variable

- Which column should be used as the target for prediction (e.g., campaign success, response rate, purchase)?

Ans: **Sales**

2.Feature Selection

- Which features (columns) are most relevant for predicting the target variable?

Ans: **Total Leads, Sales Conversion Rate (%), Cost per Lead and Total Cost**

3.Model Selection

- Which ML algorithm would you use — Logistic Regression, Decision Tree, or Random Forest — and why?

Ans: Target is numeric and continuous, so it is a regression problem. I will choose Decision Tree Regressor or Random Forest Regressor.

4.Data Splitting

- How would you split the data into training and testing sets (e.g., 80/20 or 70/30)?

Ans: test size=0.30 so training=70% and testing=30%

5.Model Training

- How will you train your model to predict campaign success?

Ans: To train the model to predict campaign success, I first define **Sales** as the target variable, since it represents the outcome of a campaign. I then select relevant features such as **Total Leads, Sales Conversion Rate (%), Cost per Lead Total Cost** which influence sales performance. I split the dataset into **training and testing sets** to evaluate the model. I use a **Random Forest Regressor** to train the model because it can capture non-linear relationships. After training, I evaluate the model using metrics such as **R^2 and error measures**..

- What performance metrics (accuracy, precision, recall, F1 score) will you use to evaluate the model?

Ans: I evaluate the model using metrics such as **R^2 and error measures**

6.Model Optimization

- How can you improve model performance (e.g., hyperparameter tuning, feature scaling, or handling class imbalance)?

Ans: Model performance can be improved through hyperparameter tuning using GridSearchCV, selecting relevant features to avoid redundancy and data leakage, and choosing an appropriate model. Since I used a Random Forest Regressor, feature scaling was not required.

Result Interpretation

- What are the top factors influencing campaign success according to your model?

Ans: The model indicates that Total Leads and Cost per Lead are the most influential factors, with Total Leads having the highest impact on campaign success.

- How can these insights help improve future campaigns?

Ans: These insights guide better budget allocation, improve lead quality, and help replicate successful campaign strategies to enhance overall performance.